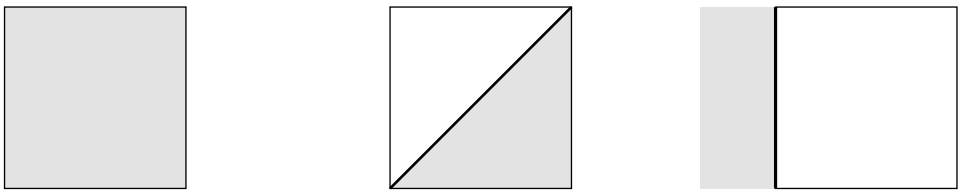
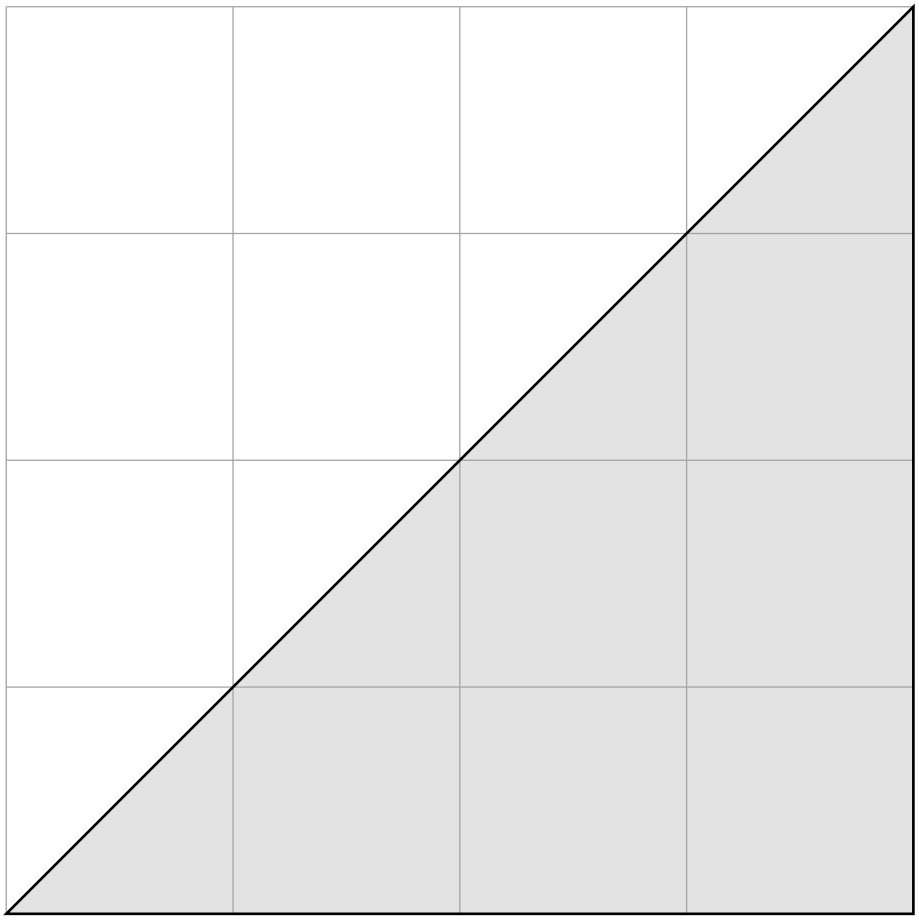
Whole squares are all gray. A cover uses every square with some gray in it. A touch at an edge or corner adds no square.



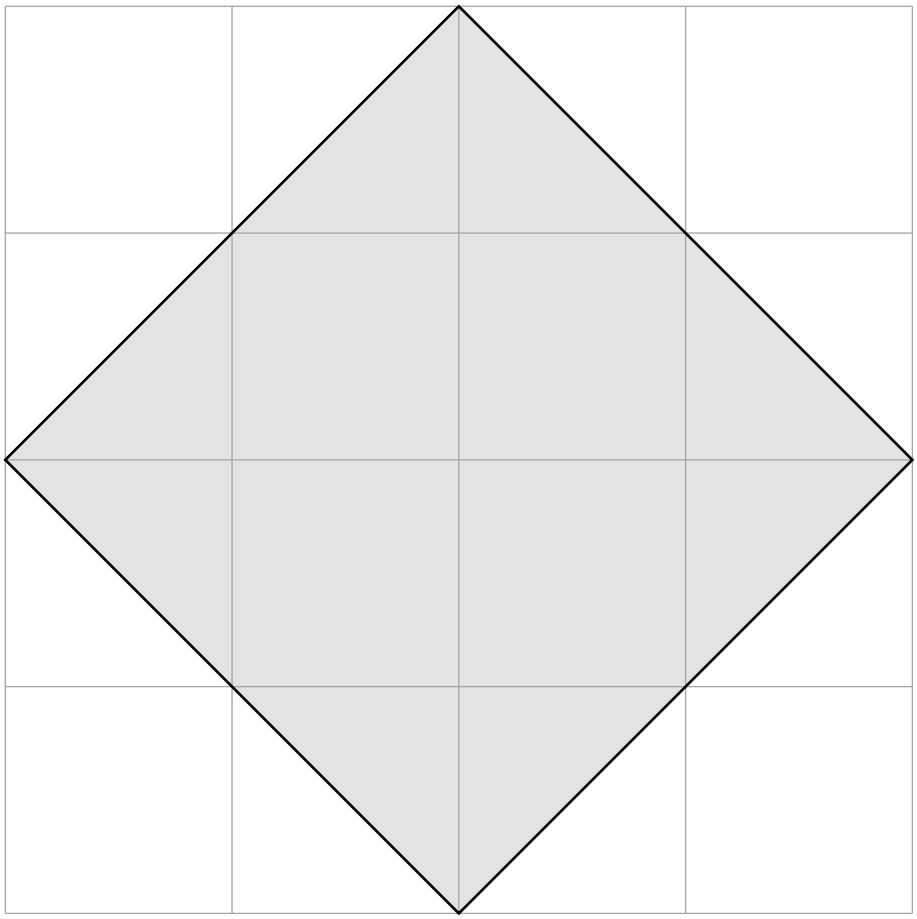
whole partial outside

Problem 1: Mark all the whole squares inside the triangle. Cover the triangle with as few grid-square tiles as you can.



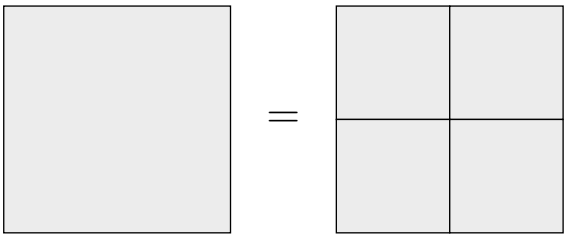
Inside: _____ big squares Cover: _____ big squares

Problem 2: Trap the diamond's area between its whole squares and its cover. Can you be certain its area is more than 3 big squares? Less than 13?



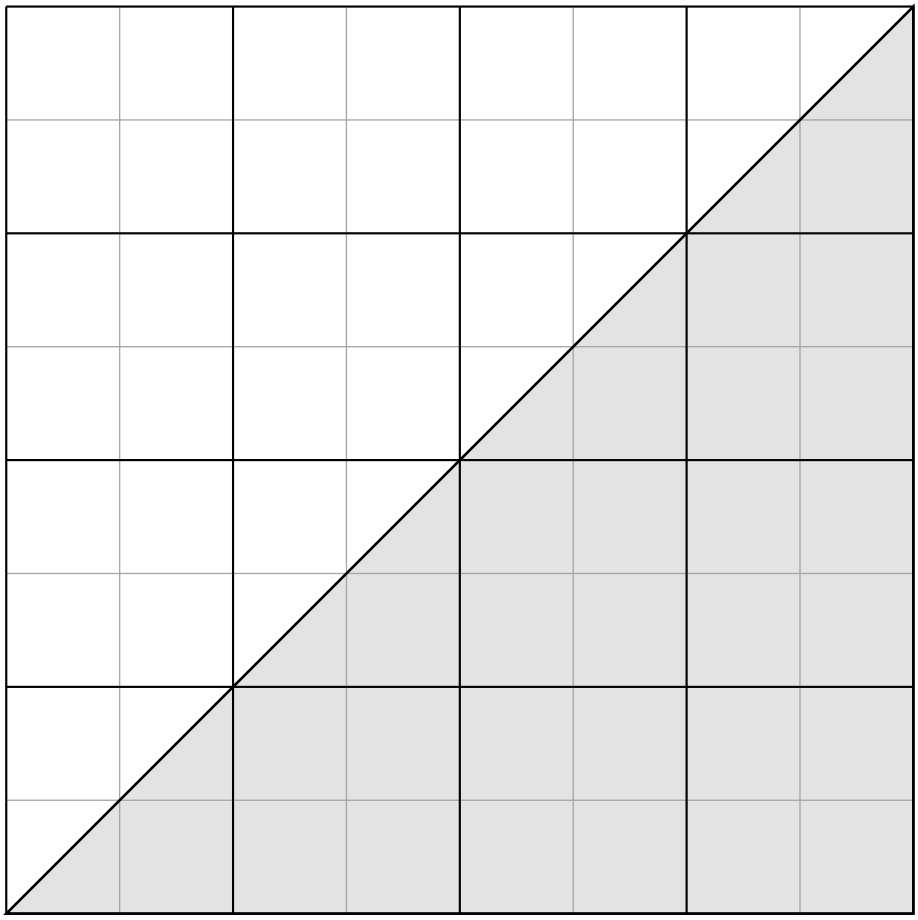
Inside: _____ big-square units

Cover: _____ big-square units



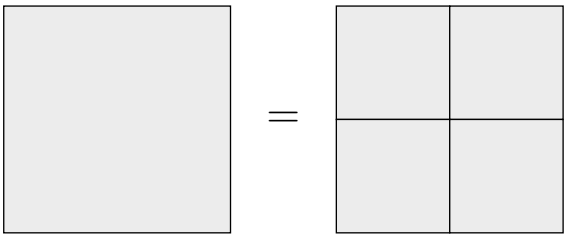
1 big square
= 4 small squares

Problem 3: Use the smaller squares to trap the triangle's area more tightly. State both bounds in big-square units.



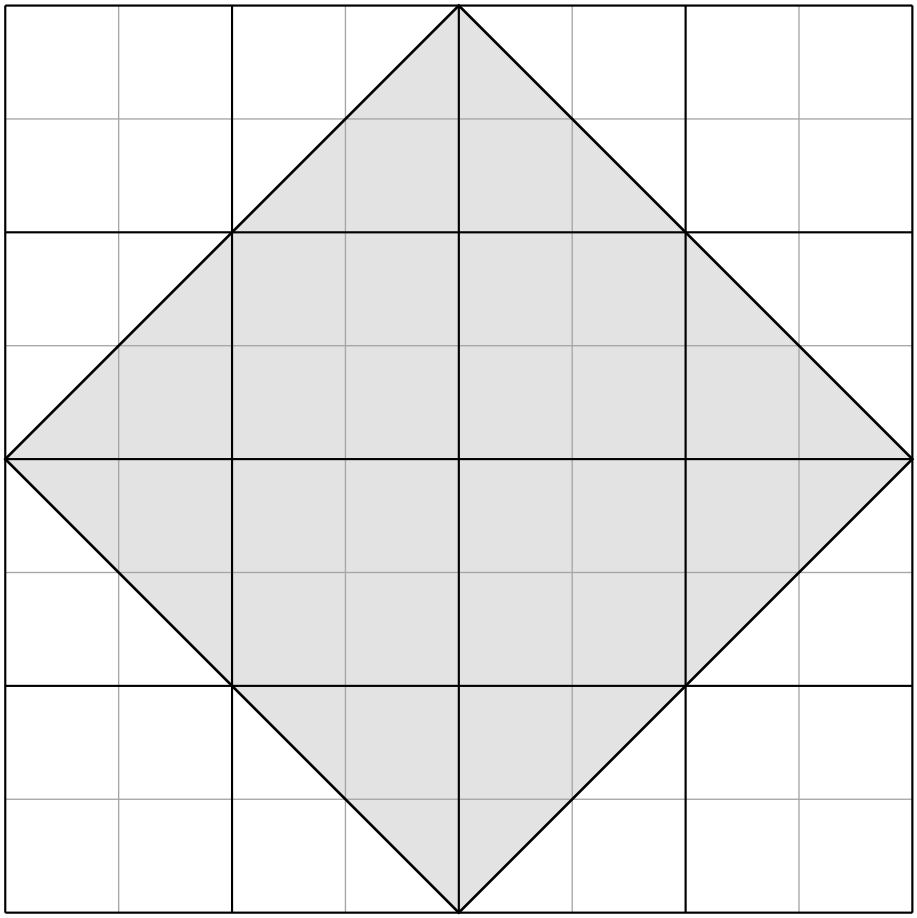
Inside: _____ big-square units

Cover: _____ big-square units



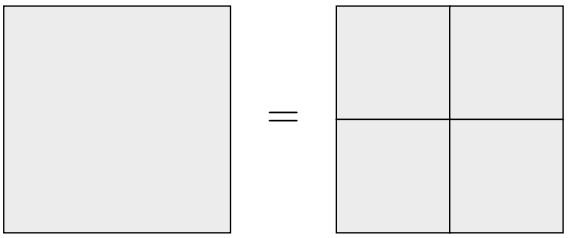
1 big square
= 4 small squares

Problem 4: Find new bounds for the diamond. Compare the area left uncertain on the big grid and on the small grid.



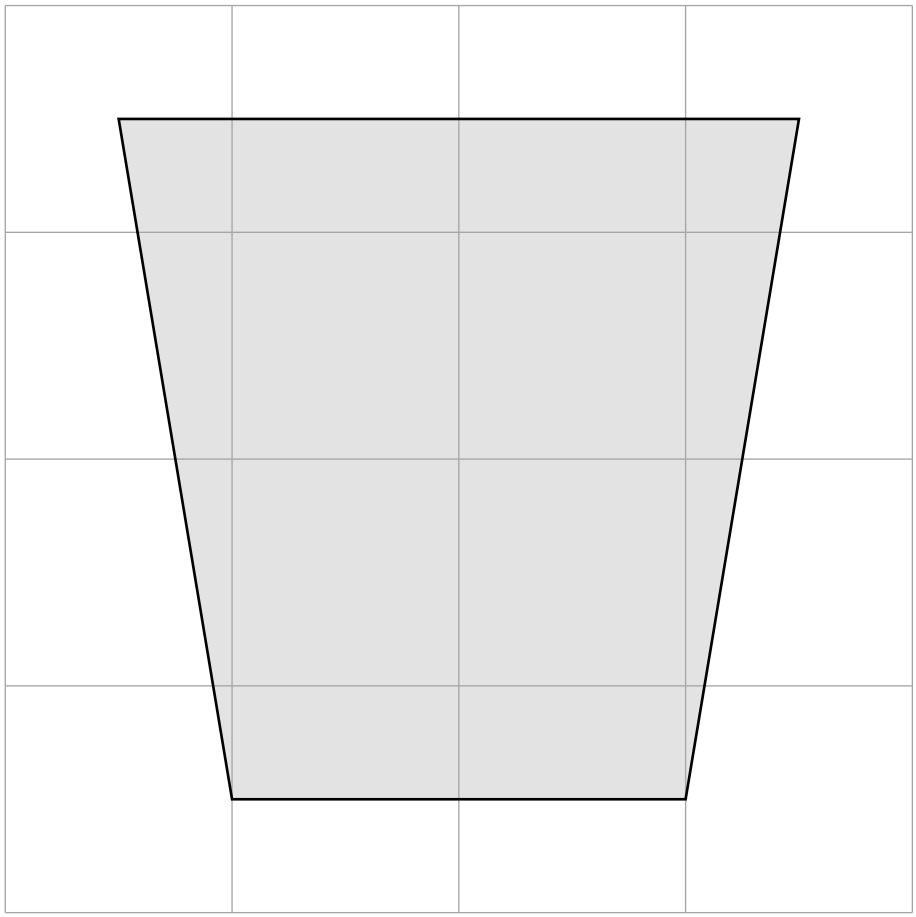
Inside: _____ big-square units

Cover: _____ big-square units



1 big square
= 4 small squares

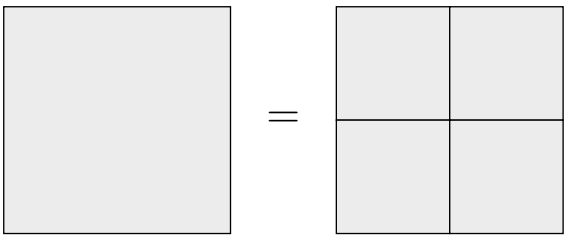
Problem 5: Split exactly three big squares into four small squares each. Choose the three that most reduce the gap between your inside count and your cover.



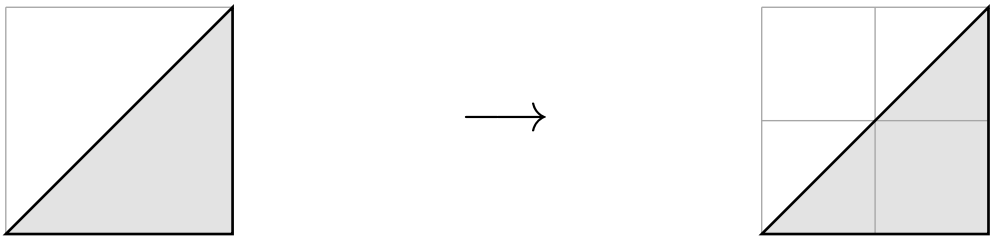
Inside: _____ big-square units

Cover: _____ big-square units

Problem 6: A big square is split into four small squares. Could the total inside area go down? Could the cover's area go up? Explain why your answers work for any of these shapes.



1 big square
= 4 small squares



Problem 7: Find the exact area of the gray shape in Problem 5. Explain how it fits between the bounds you found there.

Exact area: _____ big-square units